Political Studies (1992), XL, 38–50

The Measurement of Pure Negative Freedom

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Examining the question of whether or how far freedom is measurable contributes to the analysis of the concept of freedom in two ways. First, it involves attempting to establish criteria for answering questions about 'how free' individuals or societies are. Secondly, it helps to show how far different definitions of freedom really conflict, in as much as those definitions are themselves motivated by intuitive extent-of-freedom assessments in the first place. Critics of the 'pure negative' conception of freedom (freedom as the absence of purely physical impediments to action) have argued either that freedom is unmeasurable on such a conception, or that such a conception is counterintuitive, because the measurements of freedom implied by it conflict with the intuitive comparisons which we normally make. Closer examinations of the nature of measurement and of the nature of act individuation show both of these criticisms to be ill founded.

A Czech and a Dutchman are talking about housing problems in their respective countries.

DUTCHMAN: Housing problems we Dutch can understand, but what must be so terrible for you is not having freedom of speech to complain about them.

CZECH: But we do have freedom of speech!

DUTCHMAN: What do you mean?

CZECH: We are free to say absolutely anything we like. The only difference is that we don't have freedom *after* speech![1]

Whether or how far it is possible to measure freedom is an important yet neglected question. The answer will of course depend on how 'freedom' is defined. This paper attempts to provide an answer on the basis of the 'pure negative' conception of freedom, the conception of freedom most famously (and infamously) defended by Thomas Hobbes,[2] and most extensively defended recently by Hillel Steiner.[3] According to the pure negative definition of freedom, an individual is free to the extent that he is not *physically prevented* (intentionally

* The author wishes to thank Richard Arneson, Steven Lukes, David Miller, Hillel Steiner and Philippe Van Parijs for their comments on earlier drafts of this paper, and Pedro Lains for his advice on the subject of 'freedometrics'.

[1] Steven Lukes and Itzhak Galnoor, *No Laughing Matter: a Collection of Political Jokes* (Harmondsworth Penguin Books, 1987), p. 126.

[2] Thomas Hobbes, *Leviathan*, Ch. XXI.

[3] Hillel Steiner, 'Individual liberty', *Proceedings of the Aristotelian Society*, 75 (1974–5), 33–50; 'How free: computing personal liberty', in A. Phillips-Griffiths (ed.), *Of Liberty* (London, Cambridge University Press, 1983) pp. 73–89; *An Essay on Rights* (Oxford, Blackwell forthcoming).

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or unintentionally by other humans) from performing actions. An action is physically prevented by the existence of a physical barrier, or by the fact that one would be erected should the individual in question *try* to perform that action. A physical barrier prevents an action either because it is a direct impediment to the movement which that action would bring about, or because it prevents access to the means to bringing about such a movement (such as machinery or land). Coercion of the will (by threats or offers, for example) does not qualify as a restriction of pure negative freedom.[4] Neither is the desirability or 'significance' of an available (taken hereinafter to mean unprevented) action to be seen as relevant to the extent to which it contributes to an individual's freedom.

Steiner has himself proposed a formula for the measurement of an individual's pure negative freedom. The extent of Red's freedom, he says, is equal to the value of

$$\frac{F_r}{F_r + U_r}$$

where F_r and U_r stand for the numbers of actions Red is free and unfree to perform respectively.[5] It will be argued here that while certain objections to this formula stand, insufficient attention either to measurement theory or to action theory (or to both) has led some critics to draw hasty conclusions about the intuitive unacceptability of pure negative freedom and about the logical impossibility of its measurement. It *is* logically possible to produce measurements of pure negative freedom which will be intuitively acceptable to many negative libertarians (although not necessarily to any positive libertarians). It may well be that actual attempts at measurement would encounter further *practical* difficulties. Nevertheless, to admit such difficulties is not to deny the existence of a model towards which the freedom measurer can aspire; neither is it to claim that the intuitive unacceptability of an actually enacted measurement is not a result merely of such practical difficulties, rather than of the fact that it is based on a conception of freedom which systematically distorts the intuitive comparisons which we normally make.

While it would be nice to be able to say 'Here is the formula, now go out into the world and measure freedom', the *fiirst* object of a formula like Steiner's is to specify exactly what (if anything) is *meant* by the claim that Red is 'very unfree', or the claim that Red is 'freer' than Blue, or even only the claim that Red's freedom has 'increased' or 'decreased'. This paper is concerned primarily with that first object. One of its main motivations is the idea that examination of the problems of freedom measurement may help us to achieve consensus over the definition of freedom – where before, consensus was thought to be impossible – given the fact that it is often intuitive extent-of-freedom assessments which lie behind disputes about the definition of freedom in the first place. If a definition of freedom which you oppose can nevertheless be shown to give rise to the kind of measurements of freedom with which you agree, do you *really* need to oppose that definition?[6]

[4] A threat is *inter alia* a *description* of a reduction in pure negative freedom (i.e., not such a reduction *per se*). The truth of that description depends on whether or not that reduction in freedom would be imposed, given the threatened agent's failure to comply.

[5] Steiner, 'How free', p. 74.

[6] Likewise, and as we shall see, a persuasive tactic often implicitly used in criticizing a certain

The remainder of this paper argues for three claims: first, that the question of whether or how far freedom is measurable is an important one for contemporary political philosophers; secondly, that it is possible to measure pure negative freedom; and thirdly, that resultant measurements of pure negative freedom need not be as counterintuitive as might first appear. A final section will contain an improved formula which is intended to accommodate the arguments put forward in support of the second and third of the above claims.

Why Measure Freedom?

Even before we begin to assess the degree to which freedom is measurable, we come across an immediate objection: that while it may be interesting to 'compare' societies and individuals in terms of their freedoms, there is no need to introduce talk about the '*measurement*' of freedom. This objection is usually based on no more than a squeamish (and unreflective) reaction to the use of the word 'measurement' in the context of political philosophy. If you are not interested in the possibility of measuring freedom, then freedom cannot (for you) be more than a quality which certain people do or do not possess in certain situations. For if you were to be interested not only in the truth value of statements asserting that an individual 'is or is not free' to perform a certain action, but also in the truth value of statements asserting 'how free' – either absolutely *or* comparatively – that individual is (in some overall sense, with regard to some specified set of available and unavailable actions[7]), then you would be assuming freedom to be a *quantitative attribute*. And a quantitative attribute is one which can be measured; it is one which can be correctly shown not only to be possessed or not possessed by an object but also to be possessed to a greater or lesser degree.[8]

Truth, death and pregnancy are not usually treated as quantitative attributes; heat, utility and intelligence usually are. When an attribute is assumed to be a quantitative one (an assumption which, if it is correct, will allow for greater predictive, descriptive and prescriptive powers), it becomes crucial to know whether our powers of measurement can meet the demands made by such an assumption (the demands being that we be able to predict, describe and prescribe in certain ways). Hence we have the vast quantity of work by physicists, economists and psychologists on the possibility of measuring heat, utility and intelligence. Work on the possibility of measuring freedom would not be called for if political philosophers did not continually assume freedom to be a quantitative attribute. But they do, and what is more, the predictions, descriptions and prescriptions which depend on that assumption are central to many of their theories.

First, freedom is assumed to be a quantitative attribute by any principle of equal or maximal freedom. The principles of equal and maximal freedom, widely employed by liberal and socialist writers, are most influentially combined in John Rawls's 'first principle of justice', which states that everyone has 'an equal right to the most extensive basic liberty compatible with a similar liberty for all'.[9]

definition of freedom is that of saying: 'If your definition of freedom can be shown to imply the kind of measurements which you find unacceptable, can you *really* endorse such a definition?'

[7] Few people would wish to claim that an individual 'is or is not free' in an overall sense; i.e., that he is either completely free or completely unfree.

[8] M. R. Cohen and E. Nagel, *An Introduction to Logic and Scientific Method* (London, Routledge and Kegan Paul, 1934), pp. 289–91.

[9] J. Rawls, *A Theory of Justice* (Cambridge, MA, Harvard University Press, 1971), p. 60. See

Now it might be pointed out that Rawls's 'basic liberties' are only formal in nature, and that the measurement problems involved in 'trading off' losses in liberty for greater gains in liberty are largely avoided by his distinction between liberty and 'the worth of liberty'; that is, between liberty and 'the value to individuals [given their resources, capabilities, etc.] of the rights that the first principle defines'.[10] However, this distinction is not even one which most negative libertarians would accept. Most would say, for example, that the *amount* of property I possess, rather than merely the right 'to hold [an unspecified amount of] (personal) property'[11] affects how much liberty I have. Since Rawls nevertheless puts great emphasis on 'the fair *value* of equal liberties' enforced by the difference principle,[12] he must also assume the theoretical possibility of what many would call 'maximal equal liberty', only under the alternative name of 'the maximal equal worth of liberty'.[13]

Secondly, any generalized claim about the relationship between freedom and egalitarian or historical entitlement conceptions of justice assumes freedom to be a quantitative attribute. Robert Nozick claims that 'liberty upsets patterns', none of which 'can be continuously realized without continuous interference with people's lives'.[14] If this claim is to have any weight (if we are not to be able simply to answer that patterns also preserve liberty), then Nozick must be taken to be saying something like this: 'If you allow *more* liberty your favourite pattern will be upset; if you want to keep your favourite pattern, you must allow *less* liberty'. G. A. Cohen, the foremost of Nozick's Marxist critics, has pointed out that if we reject Nozick's 'moralized' definition of freedom (whereby violations of Nozickian rights are the only preventative actions which count as restrictions on freedom) and admit that 'private property is a distribution of freedom *and* unfreedom',[15] then it will no longer be clear whether greater individual liberty is guaranteed by capitalist or by socialist structures of property rights.[16]

Why Measure Pure Negative Freedom?

It is worth noting that the move from a Nozickian to a Steinerian definition of freedom is sufficient to make the above issue a contentious one. As a result, both the right and the left stand to gain much if they can prove their point on the basis of the pure negative conception of freedom. (This is not, of course, the same as saying that they should become pure negative libertarians.) More left-wingers will be swayed by right-wing arguments which do not rely on a moralized definition of freedom, while left-wingers will themselves sway more right-wingers if they do not rely on the introduction of those non-libertarian premises which imply a more positive definition of freedom. Ernest Loevinsohn, for example, has

also p. 244, where Rawls prescribes that 'liberty . . . be restricted only for the sake of liberty itself', thus implying that 'if liberty is less extensive, the representative citizen must find this a gain for his freedom on balance'.

[10] Rawls, *Theory of Justice*, p. 204.

[11] Rawls, *Theory of Justice*, p. 61.

[12] Rawls, *Theory of Justice*, p. 277.

[13] See Norman Daniels, 'Equal liberty and unequal worth of liberty', in N. Daniels (ed.), *Reading Rawls* (Oxford, Basil Blackwell, 1975), pp. 253–81, for a full analysis of this distinction.

[14] Robert Nozick, *Anarchy, State and Utopia* (New York, Basic Books, 1974), p. 160.

[15] G. A. Cohen, 'Illusions about private property and freedom', in J. Mepham and D. H. Ruben (eds), *Issues in Marxist Philosophy, Vol. 4* (Hassocks, Harvester, 1981), p. 227.

[16] Cohen, 'Illusions about private property and freedom', pp. 232–42.

put forward an argument for the enforced redistribution of resources which he claims proceeds on the basis of libertarian premises. But this is not so, because his argument depends first on the claim that the 'importance' of taxed resources is greater for the poor than for the taxed rich, and secondly on the claim that 'the more important the blocked course of action is to the person, the more the person's liberty is curtailed'.[17] Right-wing libertarians would dispute the first claim and, in any case, reject the second outright.

Assuming the pure negative conception of freedom, then, may allow for more dynamic and fruitful debates about the relationship between freedom and competing conceptions of justice. However, it is not difficult to see that a necessary condition of our successfully drawing the battle lines in this way (of the possibility of one argument having any persuasive power over another) must be agreement not only on the definition of freedom, but also on criteria for assessing the truth value of statements about its extent.

Can We Measure Pure Negative Freedom?

What, then, of Steiner's proposed criteria for such assessments? That is to say, what of his proposed formula for the measurement of pure negative freedom? Measurement is 'the assignment of numerals to objects or events according to rules'.[18] For Steiner, straightforwardly enough, the measurement of freedom consists in assigning the numeral '1' to all actions (because of the rejection of 'valuational magnitudes' as relevant to extents of freedom), and then summing up those numerals assigned to the actions available to Red, on the one hand, and those assigned to the actions unavailable to Red, on the other.

We must, however, be more clear about what is meant by 'an action'. If I can only go (am only unprevented from going) to East Berlin for one day but you can go for a year, we are both free to 'go to East Berlin'. If I can only go to East Berlin but you can go anywhere in East Germany, we are both free to 'go to East Germany'. But we do not have the same amount of freedom, for the simple reason that the extents to which we are subject to prevention are not equal. Given this, it must be that what is meant by 'an available action' is an available *act token*, which may occur only in a specific place at a specific time, rather than an available *act type*, which never has both a spatial *and* a temporal location. Had we merely counted the act types of 'going to East Berlin' and 'going to East Germany', we might have been described (incorrectly) as equally free with respect to travel behind the Iron Curtain.

It is also worth noting that if we count act types rather than act tokens, then trade-offs in freedom will not be a zero sum game, as Steiner has suggested.[19] If, for example, I possess two houses and you possess none, and one house is forcibly transferred to you, then I have not lost the possibility of 'living in a house' but you have also gained it. In other words, the transfer of property from me to you actually appears to have increased the sum total of freedom available to the two of us. It becomes obvious that we must list the specific actions 'live in house *A*'

[17] Ernest Loevinsohn, 'Liberty and the redistribution of property', *Philosophy and Public Affairs*, 6 (1976–7), p. 233.

[18] This is measurement defined 'in its broadest sense', by S. S. Stevens, in 'Mathematics, measurement and psychophysics', in S. S. Stevens (ed.), *Handbook of Experimental Psychology* (New York, Wiley, 1951), pp. 1–49.

[19] Steiner, 'Individual liberty', pp. 49–50; Steiner, 'How free', pp. 88–9.

and 'live in house *B*', and exclude the act type embracing those specifics ('live in a house') in order that our measurement represents the fact that I have suffered at least some loss in freedom. (Pure negative freedom is the 'personal possession' of the 'physical space and matter' which allows us to perform actions,[20] an example of which is a house).[21] It is obvious enough that if we 'double count' available actions, then levels of freedom will be represented by our measurements as greater than they really are according to the pure negative conception of freedom. Counting act types can lead to the double counting of act tokens on a vast scale. If I were to declare my house to be the property of all British citizens, I could, by counting available act types, claim to have added around 50 million more available actions to the list of those already available to the people of Britain, without making *any* of the actions on that list *un*available to anyone.

But the counting of act tokens would also appear to be problematic. Even when they are spatio-temporally specified (and even rejecting the relevance of 'valuational magnitudes') we will not want simply to assign identical numerals to 'travelling around the world' and 'twiddling our thumbs', such that if I can only do the former and you can only do the latter, we are said to be equally free with respect to this list of two actions. Steiner is aware of this and as a solution suggests that we break up actions into their constituent parts. The idea here is that we will find many more act descriptions which comprise the former act than comprise the latter, and that 'there can never be a reason for rejecting any proposed addition to the list'.[22]

But *how* specifically should we list the component parts? Cannot space and time (theoretically) be divided up infinitely? We are up against the objection to the possibility of measuring freedom put forward by Onora O'Neill, that 'it would always be possible to show that any given set of liberties was as numerous as any other merely by listing the component liberties more specifically'.[23] The set of actions which I am free to perform and the set of actions which I am unfree to perform would both appear to be of an arbitrary size. On one account I might be very free, on another I might be very unfree, and we would seem to be left with no acceptable criteria for judging between these two accounts. Steiner says that while the list of actions which a prisoner is prevented from doing is 'indefinitely long', 'it is also true that there is an indefinitely long list – *though not as long as the previous one* – of actions which this individual is not prevented from doing'.[24] Yet he has no grounds for saying that the second list is the shorter one. Any mathematician will point out that two infinitely large sets can never have different numbers of members (even if one of them is a subset of the other, a condition that does not in any case apply here) and we cannot therefore say (as we would like to) that the prisoner has less freedom than unfreedom.

Does this problem really show that we cannot measure pure negative freedom? Before jumping to conclusions, we should seek to know more about what it

[20] Steiner, 'Individual liberty', p. 48.

[21] Whether or not all examples of trade-offs in freedom support Steiner's 'zero-sum' conclusion, however, is open to question. Often, the actions made unavailable to the loser of freedom are *different* from the actions made available to the gainer of freedom. A government which restricts the freedom to use private health care no doubt creates other freedoms; but *how much freedom*? (I owe this example to Tim Gray.) Act-type counting, on the other hand, does *not* provide evidence against the 'zero-sum' thesis because it does not provide a 'valid' measurement of pure negative freedom (see below).

[22] Steiner, 'How free', p. 74.

[23] Onora O'Neill, 'The most extensive liberty', *Proceedings of the Aristotelian Society*, 80 (1979), p. 50.

[24] Steiner, 'Individual liberty', p. 45; emphasis added.

means to *measure*. A measurable quality is either *intensive* or *extensive*.[25] Where a quality is *extensive*, it is possible to *add* one instance of it to another. Here, the possibility of an addition, or '*concatenation*' operation, is what allows us to say *how much* of a certain quality an object possesses. Thus, for example, if using a balance we have found that object B is equal in weight to object B', and we also now find that $B + B'$ (B concatenated with B') is equal in weight to object A, then the number we assign to the weight of A should be twice that assigned to the weight of B.[26] The measurement of an extensive quality involves an *empirical counting procedure*. It is on the basis of the counting of individual units which have been demonstrated to be of equal size that we can say that one object possesses, for example, 'twice as much' of a certain extensive quality as another object. An *intensive* quality is one which cannot be measured by any straightforward empirical counting procedure. Examples of such qualities are the 'hardness' of rocks or the 'intelligence' of humans. It is a matter of dispute whether cardinal measurements of these qualities are strictly possible.[27] What is important for our purposes, however, is to know on what grounds a measurement is claimed to be *isomorphic* with the set of empirical relations it is intended to represent.[28] Only when isomorphism obtains, can a measurement be recognized as a *valid* one.[29] In the case of an extensive (but not in that of an intensive) quality, isomorphism is guaranteed by the possibility of distinguishing and concatenating individual instances of that quality.

Now it would certainly appear that one of the assumptions made by Steiner about freedom measurement is that it is the measurement of an extensive quality. Indeed, this assumption is a logical implication of the assumption that freedom measurement involves no more than the counting of empirical (spatio-temporally specified) possibilities of action. But where is the possibility of concatenation on the basis of which we are justified in assigning the numeral '1' to each action? Steiner's reason for doing so is the unacceptability for a negative libertarian of assigning different numerals to different actions according to their desirability or 'significance'. (It is unacceptable, Steiner suggests, because of the thereby implied assignment of negative numbers to certain actions.[30]) But this is a negative reason; it is a reason *against* using a certain rule which assigns different numerals to different actions, but not a reason *for* a particular rule which assigns identical numerals to all actions. (Note also that Steiner cannot answer that his reason for assigning '1' to each action is the equal value of actions, for the very reason that the extent to which actions are valued is, for him, *irrelevant* to questions about extents of freedom.) In order to count as an extensive measurement of quality x, an empirical counting procedure must not be simply the counting of instances of quality x, but rather, the counting of instances of

[25] Cohen and Nagel, *Introduction to Logic and Scientific Method*, pp. 293–7.

[26] Cohen and Nagel, *Introduction to Logic and Scientific Method*, p. 296; Cf. C. H. Coombes, R. M. Dawes and A. Tversky, *Mathematical Psychology: an Elementary Introduction* (New Jersey, Prentice Hall, 1970), pp. 10–11.

[27] This is denied by N. R. Campbell in *An Account of the Principles of Measurement and Calculation* (London, Longmans, Green, 1928).

[28] Isomorphism in measurement is 'a correspondence between an empirical and a numerical relational system'. Systems α and β are isomorphic 'if both β represents β and α represents β'. See Coombes et al., *Mathematical Psychology: an Elementary Introduction*, pp. 10–11.

[29] 'This is the question of validity: given that we have a fine reliable test, does it measure what we intended it to measure?'. See J. Kyburg, *Theory of Measurement* (Cambridge, Cambridge University Press, 1984), p. 248.

[30] Steiner, 'How free', pp. 79–83.

quality x, each of which is recognized to be of a comparable quantity. As it stands, therefore, Steiner's formula is a failed candidate for extensive measurement, while it is not even a candidate for intensive measurement, implying as it does, that an empirical counting procedure is sufficient for measurement. Steiner's intuition is that if I can perform 'five actions' and you can perform 'five actions', then there is no reason for saying that we are not equally free. But this is not a reason for saying that we *are* equally free, just as having no reason for saying that God does not exist is not a reason for saying that he does.

Nevertheless, this problem has been exploited by the critics of the pure negative conception of freedom, where it has been supposed (wrongly) to give rise to counterintuitive examples of measurement. A much cited example devised by Charles Taylor compares the unfreedom incurred by bans on religious worship with that incurred by regulative traffic lights. 'In sheer quantitative terms', Taylor writes, 'the number of acts restricted by traffic lights must be greater than that restricted by a ban on public religious practice. So if Great Britain is considered a free society [despite traffic lights] why not [traffic-light-free] Albania?'[31] The answer to this question is that as far as our discussion of the nature of action-counting has gone (and it has gone further than Taylor's), to claim that traffic lights restrict 'a greater number of acts' is completely arbitrary. Even if we ignore the 'great number' of acts made available to others when I am forced to stop my car, it is no argument simply to point out that 'those who do practice a religion generally do so on one day of the week, while they are held up at traffic lights every day'. Whether or not the pure negative conception of freedom is a 'non-starter' (as Taylor claims), this has not yet been shown to be so by any logically implied 'diabolical defence of Albania as a free country'.[32] *No* valid measurements of pure negative freedom have yet been shown to be logically implied.

In contrast to Taylor, Richard J. Arneson has argued that the 'neutral counting' of available actions logically implies the claim that everyone is *equally* free and has attempted to use this counterintuitive implication in order to demonstrate the relevance of desire to the measurement of freedom.[33] In Arneson's example, Jones is at present a prisoner in a straightjacket, who can move no part of his body but his thumbs. Smith used to be a prisoner but is still suffering the disabling effects of thumb-screw torture. (Note here, that the set of Jones's available actions is not, even if described in terms of available act types, a subset of Smith's.) We all share the immediate intuition that Smith is at present freer than Jones. Because 'the "objective" count [of available actions] oddly yields a tie',[34] Arneson says, we are forced to admit that it is the *desires* of Smith and Jones (at least, the desires which we here assume Smith and Jones to have) which lie behind our judgement of their relative extents of freedom. While the ' "objective" count' may yield a tie, it is, as we have seen, not a valid measurement of freedom, and the pure negative libertarian is *not* therefore forced to say that Smith and Jones's freedom is 'on a par'.[35] As before, not to have a reason for saying that Smith and Jones are not equally free, is not to be forced to say that they are.

Of course, the pure negative libertarian has yet to show that he *can* measure freedom. For this purpose, a second example of Arneson's is particularly helpful,

[31] Charles Taylor, 'What's wrong with negative liberty?', in A. Ryan (ed.), *The Idea of Freedom* (London, Oxford University Press, 1979), p. 183.

[32] Taylor, 'What's wrong with negative liberty?', p. 183.

[33] Richard J. Arneson, 'Freedom and desire', *Canadian Journal of Philosophy*, 3 (1985), 425–48.

[34] Arneson, 'Freedom and desire', p. 443.

[35] Arneson, 'Freedom and desire', p. 427.

because it shows up the real difference between the two intuitions which motivate most negative conceptions of freedom on the one hand and Arneson's 'desire thesis' on the other.[36] In order to see that it is the desire thesis which sways our judgement about Smith and Jones's freedom, Arneson argues, we should imagine the situation altered, such that 'both Smith and Jones are fanatical adherents of an obscure religion which places overwhelming value on the performance each day of a great array of devotional exercises performed with the thumbs'.[37] Unfortunately for Arneson, the average negative libertarian still does not hesitate in his intuitive judgement (this time contrary to Arneson's) that Smith is freer than Jones. Nevertheless, Arneson does believe the introduction of his second example to add some intuitive weight to his argument. Why is this? The only difference in the extent to which the agents' desires are satisfiable in the two examples is that in the first Smith's are more satisfiable and in the second Jones's are more satisfiable. But this difference is not sufficient to make the second example more persuasive than the first. And the only other difference (which Arneson must suppose makes the second example more persuasive) is to be found in the idea that in the second example, the agent whose desires are the *more* satisfiable, is also the agent who has '*less* available action'. Arneson has therefore provided us with an alternative to the quantitative attribute 'desire', upon which to base the possibility of concatenating actions (while remaining consistent with the intuitive extent-of-freedom judgements which motivate the pure negative conception of freedom); that is, the quantitative attribute 'action'.

Action, like butter, is itself a quantitative attribute, in addition to being good or bad, desirable or undesirable. Just as any lump of butter, or set of lumps of butter, is itself a 'quantity of butter', so any one action, or set of actions, is itself a 'quantity of action'. If we take the Smith–Jones examples as a guide, it would seem reasonable to say that our notion (shared by Arneson) of 'quantities of action' is based on the spatio-temporal dimensions of the particular physical events which agents bring about. That is, the greater the difference between the spatio-temporal coordinates of a particular event, the greater the 'quantity of action' performed by an agent when he brings it about. If, on this basis, we can establish intersubjective agreement (however rough the estimates) on the truth of the claim 'if action $a = 2$, then action $b = 1$', then to divide action b (physically) into two further actions, say c and d, will not produce any difference in resultant measurements, because it will be further agreed that $c + d = b$, and thus that if $c = 0.5$, $d = 0.5$, that if $c = 0.75$, $d = 0.25$, and so on. There will therefore be no *point* in dividing up b into its constituent acts, unless c and d are two actions available to two different agents.

How Acceptable are Measurements of Pure Negative Freedom?

We have taken long enough to prove a fairly obvious point; that if freedom is the availability of physically unprevented actions, and if it is possible to measure physically unprevented actions, then it is possible to measure freedom. But it is important to establish that point if one is to argue further that such measurements need not be counterintuitive. It is just because they *will* be valid

[36] The desire thesis is: 'The amount of freedom a person possesses varies directly with the extent to which his desires (or personal values) are satisfiable under the options available to him'; see Arneson, 'Freedom and desire', p. 431.

[37] Arneson, 'Freedom and desire', p. 427.

measurements (unlike the measurements of pure negative freedom performed by Arneson and Taylor) that demonstrations of their intuitive unacceptability (or acceptability) will be seen to carry weight. Let us therefore now look at two possible objections which suggest that valid measurements of pure negative freedom will still be counterintuitive.

The first objection is the familiar claim that, for the pure negative libertarian, a society with many oppressive laws need not be seen as very unfree, since such laws leave the agent 'free' to perform (*physically* unprevented from performing) the very actions they prohibit. Similar to this is the (also familiar) claim that, for the negative libertarian, the penniless tramp is 'free' to dine at the Ritz. These two claims might be said to be objections to different things: the first to Steiner's argument that threats do not reduce freedom,[38] and the second to the negative libertarian's distinction between unfreedom and inability. But the claims are similar, in as much as they both rely for their persuasive force on a confusion of available act tokens with available act types. Goebbels was not lying when he said that 'anybody can write what he likes if he is not afraid of the concentration camp'.[39] But the pure negative libertarian does not simply say '*A* in Nazi Germany can write what he likes and *B* in Great Britain can write what he likes'. 'Writing what he likes' is an act type, of which *A* has available a much smaller quantity in terms of act tokens. (*If* he writes x at time t, he may not be allowed to do so at $t + 1$). What is more, there is a vast (spatio-temporally defined) quantity of other kinds of action which *A* may be prevented from performing *if* he writes what he likes (given that there is not 'much' to be done in a concentration camp). The same is true of the tramp. He *is* free to dine at the Ritz: once, and only once (if he does not get thrown out first for bad dress).[40]

The unfreedom of *A* and of the tramp can be represented numerically in the following way. Some possible act combinations are 'and/or' options; we can perform either one or the other or both. Others are 'or only' combinations; we can perform only one or the other. Therefore, each agent has available to him various *sets of compossible actions* (for example, in society *A* I can do xyz, xy, xz, yz, x, y or z, and in society *B*, where x is a criminal act, you can only do yz, y or z).[41] The extent to which the availability of spatio-temporally defined action x contributes to my freedom will be the number we have assigned to x, multiplied by the number of sets of compossible actions in which x appears. There is no need, then, for the negative libertarian to deny the obvious (that a law does not *itself* restrict negative freedom) merely in order to allow for intuitively acceptable measurements.[42]

[38] Steiner, 'Individual liberty'.

[39] Cited in A. and D. Gabor, 'An essay on the mathematical theory of freedom', *International Journal of Social Economics*, 6 (1979), 346.

[40] Steiner would not accept Rawls's distinction between liberty and 'the worth of liberty' (*Theory of Justice*, p. 204), in as much as the worth of liberty to do x is guaranteed by the *means* to do x (for the tramp, this is money), in the absence of which x would be physically prevented.

[41] As Morriss helpfully points out, 'the criminal law deliberately rests on the apparent paradox that we are left free to commit a crime, and *then* our freedoms are curtailed, and curtailed in a manner that may well leave us free to continue committing the crime', in P. Morriss, *Power: a Philosophical Analysis* (Manchester, Manchester University Press, 1987), p. 120.

[42] J. P. Day is mistaken in the belief that he must oppose Steiner over the relevance of threats to freedom. See J. P. Day 'Threats, offers, law, opinion and liberty', in *Liberty and Justice* (London, Croom Helm, 1987), pp. 37–42. He argues that threats restrict freedom by preventing 'complex actions (i.e., act combinations). But threats (unlike the person who makes them) can only 'prevent' such act combinations through coercion of the will (which Day does not recognize as restricting freedom). At the same time, Day would agree with the measurements produced by the pure negative

The second objection points out that despite the broad agreement of negative libertarians in opposition to Arneson (over Arneson's second example), they would nevertheless not say, for example, that the extent of freedom represented by ordinary freedoms of speech is proportional to the extent of movement of the tongue and vocal chords necessary for their exercise.

In order to answer this objection, we need to look at the way in which different actions are related. Actions are either *basic* or *non-basic*, a basic action being the bringing about by an agent of a movement of his own body, and a non-basic action being his bringing about of an event *by* moving his body.[43] Where any basic action has a direct consequence, the bringing about of that consequence also constitutes an action – a *non*-basic action. And indeed, if a basic action is spatio-temporally specified, its performance will necessarily entail the bringing about of all of its consequences; that is, a 'string' of non-basic actions. Borrowing Alvin Goldman's terminology, basic actions can be seen in this way as 'level-generating'[44] non-basic actions, which themselves in turn 'level-generate' other non-basic actions (for example, my 'moving my foot' generates my 'pressing the accelerator', which generates my 'driving my car'). All actions require certain '*physical components*' for their performance.[45] The performance of a basic action requires for its performance some part of the agent's own body and the physical space within which it moves. The performance of a non-basic action requires the agent's causal power over additional matter (the matter the movement of which that action is the bringing about, plus the physical components of any action whose performance is necessary in order to generate that action).[46]

Given this, it is obvious enough that my detention of the physical components of your doing x will take away not only your freedom to do x, but also your freedom to perform any action which would have been (and could only have been) generated by x. As Steiner suggests,[47] in order to prevent an agent from performing any actions at all, we need only deprive him of possession of his own body, because to do so is to deprive him of possession of all other physical components. The fact that most of our actions are performed *by* performing other actions would therefore suggest a certain relationship between the 'generational status' of an action and the amount of freedom which its prevention restricts: the closer a confiscated physical component is to the agent's body (in the causal chains of events of which actions are the bringing about), the greater the quantity of action its confiscation is likely to prevent.

When A physically prevents B from speaking in a certain way, therefore, there will be a great reduction in B's pure negative freedom. A will either be preventing B from performing certain bodily movements, or will at least be preventing B's access to the physical space (through which the sound waves would have

libertarian. Day should therefore admit either that coercion of the will restricts freedom, or that he is really a pure negative libertarian.

[43] See Arthur Danto, *Analytical Philosophy of Action* (Cambridge, Cambridge University Press, 1973).

[44] Alvin Goldman, *A Theory of Human Action* (New Jersey, Prentice Hall, 1970), pp. 24–8.

[45] Steiner, 'Individual liberty', p. 48.

[46] For Goldman, 'level-generation' need not be causal (e.g., 'John's extending his arm out of the window' is said to generate 'John's signalling for a turn' *conventionally*). For the purposes of the pure negative libertarian, however, two actions will be identical if their 'physical components' are identical. Causally generationally related actions will therefore be the only non-identical generationally related actions. Another way of putting this is: actions x and y are identical if to prevent x is to prevent y and to prevent y is to prevent x.

[47] Steiner, 'Individual liberty', p. 49.

travelled) which would otherwise have been directly causally linked to *B*'s body. This will also involve preventing all of the actions which would be consequent upon *B* moving his body in those prevented ways (or upon *B*'s use of the space or matter to which he has been deprived of access). Of course, laws against free speech do not usually involve such direct preventions in the first instance, but where they threaten arrests and imprisonment, the same argument applies to the incompossible set of actions, 'saying *x* at *t*' and 'walking down the street at $t + 1$', and where they threaten fines, the same argument applies to the incompossible set of actions, 'saying *x* at *t*' and 'buying a new car at $t + 1$'. (Remember that to prevent the act set 'saying *x* at *t*' and 'walking down the street at $t + 1$' is to prevent *all* the act sets which, however large, include these two actions among their members.[48])

One implication of the answers given to these two objections is that some actions are not separately performable and must not be represented in our calculations as such. Steiner is therefore wrong to say that the various act descriptions, 'he pulled the trigger of a gun', 'he fired a bullet', 'he shot a man' and 'he killed a man' can all be aggregated unproblematically in accordance with his formula, on the grounds that 'each of the . . . act descriptions . . . is only contingently related to the rest and therefore each refers to an event which could have occurred even if the others had not'.[49] Any spatio-temporally defined pulling of the trigger either will or will not give rise to a killing, and whether or not it does so will depend not on another temporally distinct action, but on the circumstances surrounding the agent at the time of that particular trigger-pulling.[50] In one possible world the trigger-pulling will occur without the killing occurring; in another it will not. But we are only interested here in a particular agent's freedom to choose between possible worlds. Although that agent *is* free to choose whether or not to kill, he is only free to do so *by* choosing whether or not to pull the trigger in a certain place at a certain time.[51]

There is indeed a sense, then, in which, as Donald Davidson has argued, *all* generationally related actions are *identical*. As Davidson puts it, if I flip a switch, turn on the light, illuminate the room and alert a prowler, 'I need not have done four things, but only one, of which four descriptions have been given'.[52] Is this a

[48] In exactly what way is a person's freedom curtailed if he is silenced by being killed (currently the Chinese government's preferred method)? This is not clear, since we commonly see dead people as neither free nor unfree. It would seem that a claim about the extent of unfreedom incurred by a killing must depend on a counterfactual claim that certain actions would have been available to the agent had he not been killed.

[49] Steiner, 'How free', p. 76.

[50] Judith Jarvis Thomson has claimed the killing to be temporally distinct from the shooting, on the grounds that a victim cannot be said to have been killed before he is dead, given, say, that he took several hours to die; see Judith Jarvis Thomson, 'The time of a killing', *Journal of Philosophy*, LXVIII (1971), 115–32. But there *can* be a temporal difference between *A*'s 'killing of *B*' (the *action* which *causes* *B*'s death) and *B*'s 'being killed' (the *state of affairs* in which *B* is dead); see L. B. Lombard, 'Actions, results and the time of a killing', *Philosophia* (Israel), 8 (1978), 341–51.

[51] This of course takes for granted that the agent has no other means (e.g., a stabbing) by which to generate that same spatio-temporally defined killing. But if he does have such an alternative means, that still does not make the trigger-pulling, the killing and the stabbing all separately performable actions. *If* the killing is performed separately from the trigger-pulling, then it is not performed separately from the stabbing. Some act sets are – because not only physically but also *logically* unperformable – neither compossible nor incompossible from the point of view of pure negative freedom. (A logically unperformable act combination is not one which it can ever make sense to talk about someone *physically preventing*.) An example of such a set would be the set of which the killing (or any other *non*-basic action) is the only member, or a set which would involve the agent being in two different places at the same time.

[52] Donald Davidson, *Essays on Actions and Events* (Oxford, Clarendon Press, 1980), p. 4.

problem for the freedom measurer? Does it mean that in order to avoid double counting, he must count only available basic actions, and thus be forced back into the position of saying that free speech counts for very little freedom? It does not. Generationally related actions are *components* of the same Davidsonian action, and the number assigned to any single Davidsonian action should therefore be equal to the sum of the numbers assigned to those individual components. Given this, counting generationally related actions as numerically distinct will not lead to double counting, as long as it is recognized that *two generationally related actions must always be members of the same compossible (or incompossible) sets*. (That is, where x and y are generationally related actions, x's membership of set A implies y's membership, and x's exclusion from set A implies y's exclusion.)

An Improved Formula

Steiner's formula does not stand up to all of the objections examined in this paper. Nevertheless, a revised Steinerian formula should do so. According to this revised formula, for any list of n actions, the extent of Red's freedom is equal to the value of

$$\frac{\sum_{i=1}^{n} A_i C_{r,i}}{\sum_{i=1}^{n} A_i C_{r,i} + \sum_{i=1}^{n} A_i I_{r,i}}$$

where A_i stands for the numeral we have assigned to an act token, i, $C_{r,i}$ stands for the number of sets of compossible actions available to Red, of which i is a member, and $I_{r,i}$ stands for the number of sets of actions incompossible for Red, of which i is a member.

The foregoing analysis will be labelled 'reductionist' and 'philistine' by those who object outright to any such attempt to describe human freedom in mathematical terms. Unfortunately for them, however, verifying extent-of-freedom assessments involves measurement, and measurement involves mathematics. They could, of course, simply reject the idea of extent-of-freedom assessments, but that is something which many political philosophies can ill afford to do. It may be that the revised Steinerian model *can* still be shown to be inadequate with respect to the demands made by political philosophers on the concept of freedom. But it is also possible to be deceived by the immediate intuition that this is the case. Such inadequacies must be demonstrated rather than merely affirmed.